

Scalability & Future Plan

Date	2 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

The **SmartLender – Loan Prediction System** is designed to provide accurate and efficient loan approval predictions through a machine learning-based web application. Although the current system meets the project requirements, there are opportunities to improve its scalability, performance, and security. Future enhancements will focus on supporting more users, integrating real-time financial data, and expanding the application's capabilities to meet real-world banking needs.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Uses a single dataset for training	May reduce prediction accuracy for diverse real-world data	High
2	Runs only on a local server	Limits access for multiple users	High
3	Basic security and authentication	Suitable only for project demonstration	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of local users	Deploy the application on a cloud platform to support multiple users simultaneously.
2	Data Storage	Uses a CSV dataset	Integrate a relational database such as MySQL or PostgreSQL for efficient data management.

3	Performance	Predicts using a locally stored model	Optimize the application and deploy on a high-performance server for faster predictions.
4	Security	Basic input validation	Implement user authentication, data encryption, and secure API communication.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate a database and user login system	Next 3 Months	Secure storage and personalized user access
Phase 3	Add real-time credit score verification and banking API integration	Next 6 Months	Improve prediction accuracy and automate verification
Phase 4	Deploy the application on the cloud with a responsive mobile interface	Next 12 Months	Support large-scale users and improve accessibility